

Gadolinium

Gd (Gd I). Strongest classified transitions below 2 μm (NIST): 21 levels, 19 lines.

Gd ⁺ ionisation limit 49601.45 cm^{−1} = 6.14980 eV (201.607 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle U||er||J'\rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

